



East West University Course Outline

Department of Mathematical and Physical Sciences

Semester: **Summer 2025**

Course Title: Differential Equations and Special Functions **Course Code:** MAT102

Pre-Requisite: MAT101

Credit Hour: 3

Course Instructor: Trishna Datta, Lecturer, Dept. of MPS

Office Room: Room# FUB-909

E-mail Address: trishna.datta@ewubd.edu

Course Contents

1. Ordinary Differential Equations

Introduction to Differential Equations, First-Order Differential Equations, Modeling with First –Order Differential Equations, Higher-Order Differential Equations, Modeling with Higher-Order Differential Equations.

2. Partial Differential Equations

Linear Partial Differential Equations of Order One, Linear Partial Differential Equations with Constants Coefficients, Partial Differential Equations and Fourier series.

3. Special Functions

Bessel's Differential Equation and Bessel Functions, Hermite's Differential Equation and Hermite Polynomials, Legendre differential equation and Legendre polynomials, Recurrence relations for Legendre polynomials, Spherical Harmonics.

Course Rationale

Differential equations help us to model various problems related to the engineering field. And with the help of specific methods, we can find out the solution to those problems. These fields also require knowledge about special functions. Based on these requirements this course has been designed with both differential equations and special functions.

Course Objective

The objectives of the course are to

1. give knowledge which will allow the students to progress with further complicated topics
2. give skills for the implementation of mathematical knowledge and expertise.
3. learn the usage of mathematical tool in order to facilitate understanding and visualization of mathematical problems.

Course Learning Outcomes

On completion of this course, the student will be able to

1. understand of basic concepts of the differential equations.
2. solve differential equations as well as the systems of such equations.
3. apply the knowledge of the differential equations which will enable them to analyze dynamics of the processes.
4. apply computational tools to analyze the solutions.

Textbooks

1. A First Course in Differential Equations with Modeling Applications-Dennis G. Zill.
2. Ordinary and Partial Differential Equations: M. D. Raisinghania
3. Elementary Differential Equations and Boundary Value Problems (7th edition): William E. Boyce, Richard C. DiPrima
4. College Mathematical Method (Volume 1): Md. Abdur Rahman

Assessment Tools and Mark Distribution :

Mid Term Examination	25%
Final Examination	25%
Class Test (Quiz)	10%
Short Quiz	5%
Assignment	10%
Viva Voce	5%
Presentation	10%
Class Attendance	5%
Class Performance	5%
Total	100%

Trishna Datta

(Trishna Datta)

Date: 18.06.2025